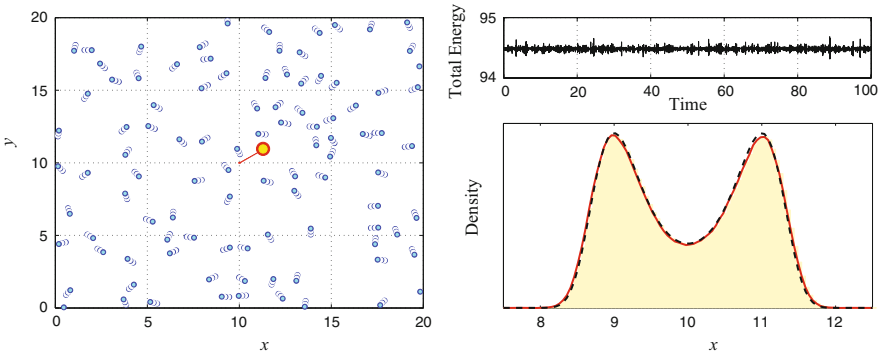


Fig. 6.2 We simulate 100 particles in the plane (*left*) using a periodic box of length $L = 20$, with the Lennard-Jones potential governing all pairwise interactions. One of these particles, marked in *red*, also feels a double-well potential in both coordinates. The total system energy (plotted in the *top right*) is conserved (with some variance due to numerical error). The distribution for the x-coordinate of the *red* particle (plotted in the *bottom right*), averaged over a single long trajectory (*solid red*) shows good agreement with the expected canonically distributed result (*dashed black*)

For example a collection of atoms on the infinite domain interacting in Lennard-Jones potentials does not satisfy the third condition, and it does not have a well defined canonical partition function. We can understand this by considering what happens over time in such a system. For simplicity, just take the case of a one degree-of-freedom system with $H = p^2/2 + q^{-12} - q^{-6}$, $q > 0$. If the temperature is low, the particle will remain for a long time in the vicinity of the well bottom. At any given time there is a positive probability of finding the particle essentially any distance from the origin, although the probability of being found very near the origin is extremely small due to the repulsive steep wall. Due to the presence of the thermal bath, eventually the particle will receive enough energy to overcome the attractive force and will be ejected and tend to infinity. The restriction to a bounded domain thus eliminates this problem which otherwise would complicate the technical treatment.

In order to use a one degree-of-freedom model on an infinite domain in the canonical setting, we would need to incorporate a restraining potential, slightly modifying for example the Lennard-Jones potential to $U(q) = q^{-12} - q^{-6} + \varepsilon q^2$ (Fig. 6.2).

6.1.4 The Canonical Density and Thermodynamic Averages

The distributional approach described above and in the previous chapter and used to define microcanonical averages on a bounded energy surface must be modified to make it applicable to the canonical ensemble. Recall that in the microcanonical setting we assumed ρ was a smooth density defined on the bounded energy surface